

Navigating subjectivity in causal context

A non-deterministic approach for information behaviour research

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The generalization impasse in LIS

- Direction: fit statistics cannot decide whether information behaviour causes culture, or culture enables behaviour.
- Logic: average effects do not show whether a context condition is a non-compensatory bottleneck.
- Aggregation: population estimates can hide the specific person, team, library, or institutional configuration that matters.
- FSSM evaluates context-specific configurations by separating typical and probabilistic forms of sufficiency and necessity.

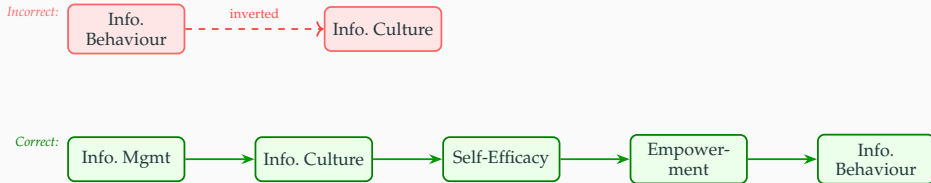
Subjective structures & effects of causes

- Subjectivity is the very area where the effect of causes is located.
- FSSM captures how the effects of a cause, whether adjacent (proximal interaction) or coercive (structural policies, culture, and norms), reside within the subjective structure of human perception rather than acting as deterministic drivers.
- It evaluates subjective structure by applying rules about effects of causes: sufficiency, necessity, adjusted probability contrasts, and case membership.
- Environmental and institutional factors do not directly force behaviour; their impact is filtered through how individuals subjectively perceive and experience affordances and constraints.

Beyond the “average” human

- Regression models produce parameter estimates: central tendencies across a population that abstract away the individual variation information behaviour research most needs to understand.
- FSSM shifts focus to subjectivity mapping, assessing:
 - Typical sufficiency: the outcome usually follows when X clears T_X .
 - Typical necessity: high outcomes rarely occur below the required level.
 - Probabilistic sufficiency: adjusted uplift when X clears T_X .
 - Probabilistic necessity: high-outcome probability when X is insufficient.
- We examine how far or close each person is from an empirically grounded threshold rather than computing aggregate summaries.

Arrow-reversal & context-first structure



- Individual sharing may explain acceptance of sharing culture, but it does not cause group information-sharing culture.
- Group norms shape individual behaviour through acculturation and coercion; information management is a common upstream condition.
- FSSM requires the structure: Management and Culture upstream; Self-Efficacy and Empowerment intermediate; Information Behaviour downstream.

The four causal claims of FSSM

1. Typical Sufficiency (α): “If $X \geq T_X$, then $Y \geq T_Y$ for most cases.” Policies usually suffice to produce the behaviour; rare exceptions are allowed.
2. Typical Necessity (ϵ): the exception rate among high-outcome cases. Low ϵ supports a non-compensatory bottleneck claim.
3. Probabilistic Sufficiency (Δ / Uplift): Does crossing T_X increase the probability of high Y under adjustment assumptions?
4. Probabilistic Necessity (β): “When X is below its bottleneck, high Y is unlikely.” Estimates high-outcome probability when the required level is not met.

Four causal claims: mathematical form

Let $b(y_i) = (T_X/T_Y)y_i$ be the case-specific bottleneck implied by the NCA ceiling.

$$\alpha = \frac{\sum_i w_i \cdot \mathbf{1}(x_i \geq T_X) \cdot \mathbf{1}(y_i \geq T_Y)}{\sum_i w_i \cdot \mathbf{1}(x_i \geq T_X)}$$

$$\varepsilon = \frac{\sum_i w_i \cdot \mathbf{1}(y_i \geq T_Y) \cdot \mathbf{1}(x_i < b(y_i))}{\sum_i w_i \cdot \mathbf{1}(y_i \geq T_Y)}$$

$$\Delta = \Pr^*(Y \geq T_Y \mid X \geq T_X) - \Pr^*(Y \geq T_Y \mid X < T_X)$$

$$\beta = \Pr^*(Y \geq T_Y \mid X < T_X)$$

T_Y is the study's outcome target. T_X is the NCA-derived bottleneck for that target. ε is an exception rate, so lower values support necessity. $\Pr^*$ denotes the weighted pseudo-population and requires explicit identification assumptions.

Dual logic: latent vs. composite route

- Route A (Latent Variable Route, PLS-SEM + NCA): Extends latent structural models. It supports population-level modelling through structural paths while attaching necessity bottlenecks and case-level diagnostics under explicit assumptions.
- Route B (Raw Composite Route, NCA + Case Weighting): Bypasses latent modelling by using raw composites. Enables a case-level “finding people” approach: NCA identifies the threshold T_X for a specific desired T_Y , and every respondent is located on the resulting membership scale.
- Route B allows researchers to study teams, even small ones, to diagnose their specific information needs and behaviours based on local evidence, without demanding full latent measurement.

Double subjectivity: entropy weighting & IPW

FSSM addresses subjectivity at two levels:

1. Measurement subjectivity: respondents read items through disciplinary lenses. Entropy weighting up-weights indicators with more discriminating information when forming construct scores:

$$E_j = -\frac{1}{\log K} \sum_k p_{jk} \log p_{jk}, \quad d_j = 1 - E_j, \quad w_j^{\text{EW}} = \frac{d_j}{\sum_l d_l}$$

2. Confounding: group differences may reflect discipline or seniority, not the construct. IPW re-balances the sample:

$$w_i^{\text{IPW}} = \frac{D_i}{\hat{e}(c_i)} + \frac{1 - D_i}{1 - \hat{e}(c_i)}, \quad \hat{e}(c_i) = \Pr(D_i = 1 \mid c_i)$$

- Together, these layers support adjusted threshold contrasts under measured-confounding, overlap, and model-specification assumptions.

Fuzzy concept & subjectivity mapping

- Construct scores are rescaled to 0–100 and mapped onto fuzzy membership intervals relative to empirical thresholds and bottlenecks.
- $\mu_N(x, y)$: degree to which an individual case meets the case-specific necessary context bottleneck.
- $\mu_S(x)$: degree to which the condition exceeds the target bottleneck for sufficiency contrasts.
- $\mu_Y(y)$: degree of outcome achievement relative to the desired target T_Y .
- Graded membership preserves case heterogeneity: it shows who clears the threshold, who falls short, and who represents a critical exception, without binary cutoffs.

Fuzzy membership: mathematical form

On 0–100 construct scores, with $b(y_i) = (T_X/T_Y)y_i$:

$$\mu_S(x_i) = \max\left(0, \min\left(1, \frac{x_i - T_X}{100 - T_X}\right)\right) \quad \mu_N(x_i, y_i) = \min\left(1, \frac{x_i}{b(y_i)}\right)$$

$$\mu_Y(y_i) = \min\left(1, \frac{y_i}{T_Y}\right)$$

The pair $(\mu_N(x_i, y_i), \mu_Y(y_i))$ locates every respondent on a context-threshold plane. The institutional environment is an ongoing context continuously shaping each individual's subjective structure. Threshold-seeking replaces point-treatment logic: not “did an intervention occur?” but “how far is this person from the enabling threshold?”

Thank you

Thank you for your attention

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FSSM Project & Updates on OSF:

`osf.io/b8maq/`